

DANIEL RAMOS

SOFTWARE ENGINEER | XR & INTERACTIVE SYSTEMS

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EXPERIENCES

Apprentice Backend Developer | Sky Portugal

Jul 2025 - Aug 2025

- Designed and implemented anomaly detection models to identify potential piracy patterns in logs from Peacock.
- Evaluated multiple unsupervised learning techniques, including DBSCAN, Isolation Forest, and Autoencoders.
- Developed a batching solution capable of evaluating 10 million log entries per run within 2-3 minutes.
- Created a streamlined interface with filtering capabilities to support efficient exploration of the results obtained.
- Delivered a solution adopted by the piracy analysis team, enabling detection of patterns difficult to identify manually.

Tech Stack: Python, Elasticsearch, ScikitLearn, Tensorflow, NumPy, Streamlit

VR Development Intern | Realizasom

Feb 2024 - Aug 2024

- Sole developer of a Mixed Reality museum experience for Meta Quest 3, currently deployed and in active use at [Ciclo da Truta de Góis - Parque da Monteiro](#).
- Built a spatial AR system that overlays interactive points of interest onto a physical map, allowing visitors to explore 360° videos and images of regional landmarks.
- Delivered the full application independently in Unity/C# with OpenXR, with a special focus on accesibility optimization for public exhibitions.
- Obtained a grade of **18** out of 20 points.

Tech Stack: Unity, C#, OpenXR, Fork

Software Engineer | O Seu Vizinho

Jun26 - Jul 14, 2023

- Co-developed a full-stack proof-of-concept platform enabling caregivers to register elderly users and manage credit for the use of services (rides, groceries, pharmacy).
- Built end-to-end in 2 weeks: user authentication, elderly registration and balance management, covering backend, frontend and database.
- Pitched at Nova impACT! Challenges, attracting post-competition interest from a jury member.

Tech Stack: PHP, Laravel, MySQL, XAMPP, HTML, CSS, Bootstrap

PROJECTS

Enhancing XR Experiences Through Multi-Device Asymmetric Collaboration

Currently developing

Thesis Project

- Designing and implementing a networked multiplayer mixed reality framework enabling asymmetric cooperation between Meta Quest 3 (VR) and Android tablet (AR) clients.
- Implemented spatial anchoring and cross-device coordinate alignment using Meta XR SDK and AR Foundation, solving shared world-space between heterogeneous XR platforms.
- Built a custom Go server using WebSockets and UDP with binary serialization for real-time, low-latency state synchronization across clients.
- Demonstrated a working asymmetric VR/AR experience at Expo FCT, where a tablet user spawned ingredients for a VR player to collect in their physical environment.

Tech Stack: Unity, C#, Meta XR SDK, AR Foundation, Go, WebSockets, UDP

GeoDescobrimentos Website

Personal Project

- Developed an interactive website that allows users to explore routes, captains and other information related to the portuguese discoveries.
- Built front-end, backend and database optimized to support geospatial data including coordinates and polygons for maps.

Tech Stack: PHP, QGIS, XAMPP, HTML, Javascript, Turf.js, CSS, Bootstrap

EDUCATION

NOVA School of Science and Technology

BSc Computer Science | Grade: 16 | 2021 - 2024

MSc Computer Science | Current Grade: 17 | 2024 - Present

Major: Interactive Multimodal Systems

SKILLS

Languages C#, Python, Go, Java, C, Javascript

Backend & Systems WebSockets, UDP, REST APIs, Elasticsearch

Tools & Infrastructure Azure, Kubernetes, Git, Linux

ML & Data Scikit-learn, TensorFlow, NumPy, Pandas